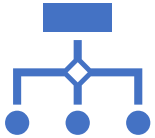
Introduction to COSC 481

Course Structure

First week: Choose project, plan project, start using GitHub for project documentation and tracking



Second week: Design and document project architecture, implement skeleton system, write tests, as time allows write commented production code passing tests

Following weeks: Write test and production code, write developer and user documentation, comment on other students' work, meet with Dr. Simon

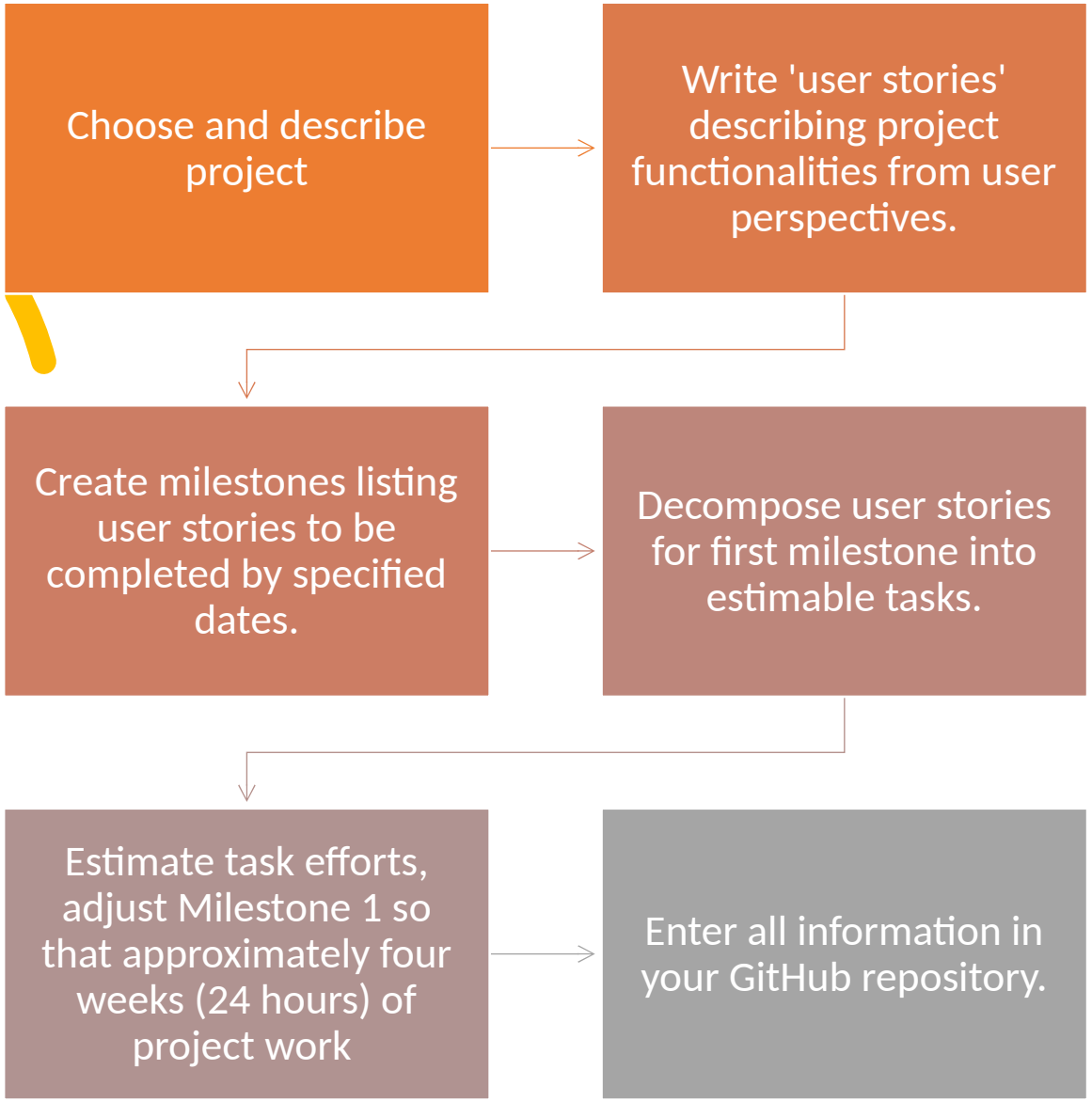
Some weeks: Present project update to the class

Grading

- Attendance 5%
- Consistent effort/progress 10%
- Quality of code, documentation, presentation, feedback provided on presentations 60%
- Achieving one or two major project milestones 0%, 15%, or 25%

But see later about effect of total hours spent on grades.

First Week Project Assignment



Choosing a Project

See syllabus for limitations, Canvas announcements for ideas.

Think big: You'll be spending a lot of time on this (average 7 hours per week)

Think small: You will be required to

- Write high-quality code
- Design and write good tests
- Include good comments conforming with Javadoc (or other language) conventions
- Write reasonably good user documentation
- Meet with Dr. Simon weekly to review your progress (plan on 30 minutes; counts toward out-of-class time)

GitHub

- GitHub is a cloud code-sharing service
  - Code sharing is based on the Git system
- You will each have a code repository within our GitHub class ‘organization’, Duquesne-Spring-2025-COSC 481
- The repositories are
  - owned by me, so that I can enforce certain policies regarding code check-in
  - "private" in the sense that they are visible only to me and your classmates
  - writeable only by the student named on the repository (and by me, but I will not be modifying your repositories)

Overview description of your project

Edit the README.md file in your GitHub repository on a "branch" to be a somewhat detailed description of your project.

You will need to “Commit” your changes when you are done editing.

Uses markdown syntax for formatting

Single line breaks are ignored. Multiple line breaks begin a new paragraph.

One or more # symbols for section headings (more #, lower level of heading)

Can style text, create bulleted or numbered lists, etc.

User Stories

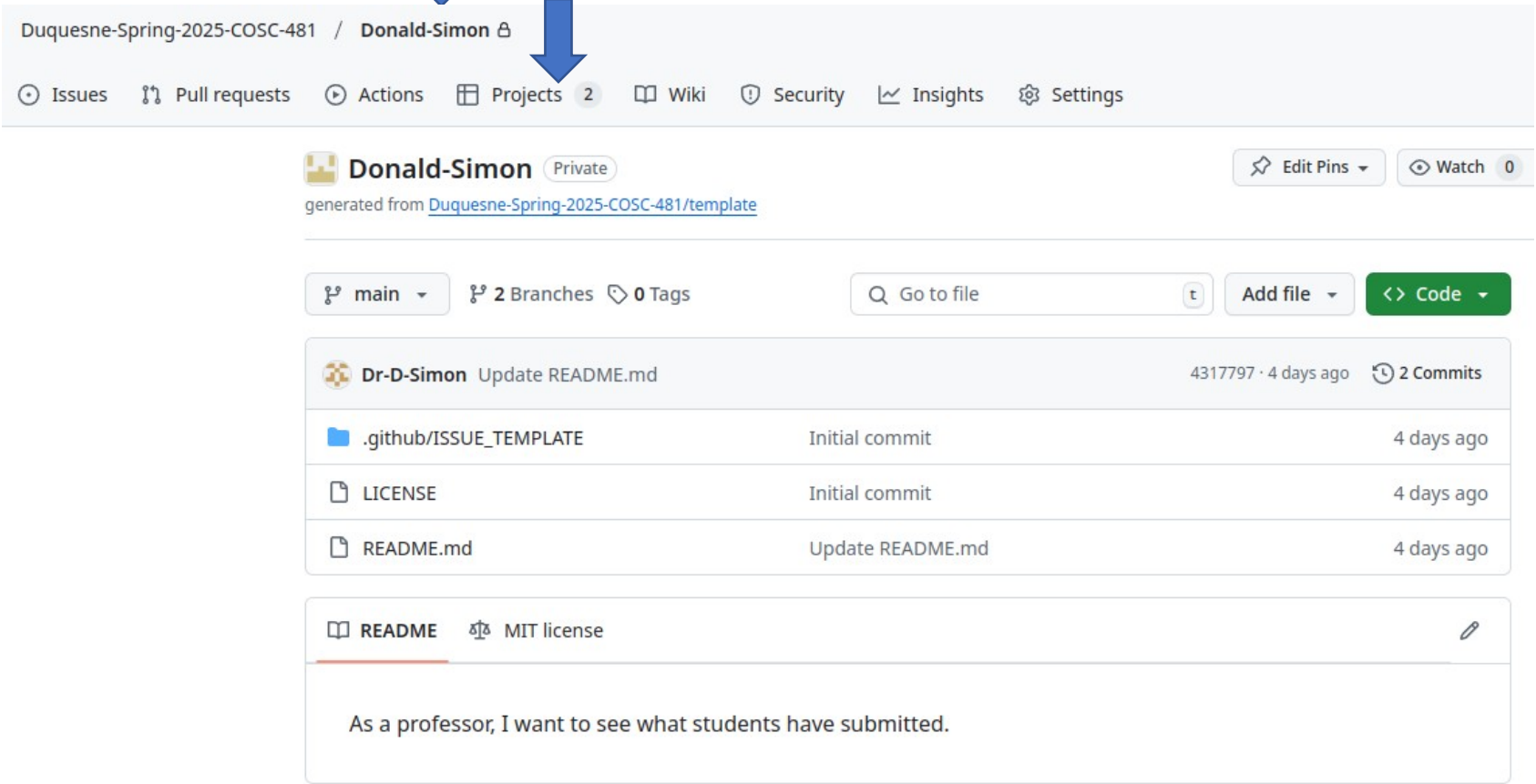
- A user story is a one-sentence (normally) description of a feature that an end user, maintainer, or other “actor” will use and the value of that feature to them
  - “As a game player, I want to be able to select an avatar so that I will be easily recognizable in the game.”
  - “As a homework-set author, I want to easily write true/false questions so that I can quickly and objectively assess student knowledge.”
  - “As a web-system maintainer, I want to be able to block denial-of-service network traffic so that we can continue to serve our clients.”

User Stories in GitHub

- Each user story will be its own **Project** in GitHub

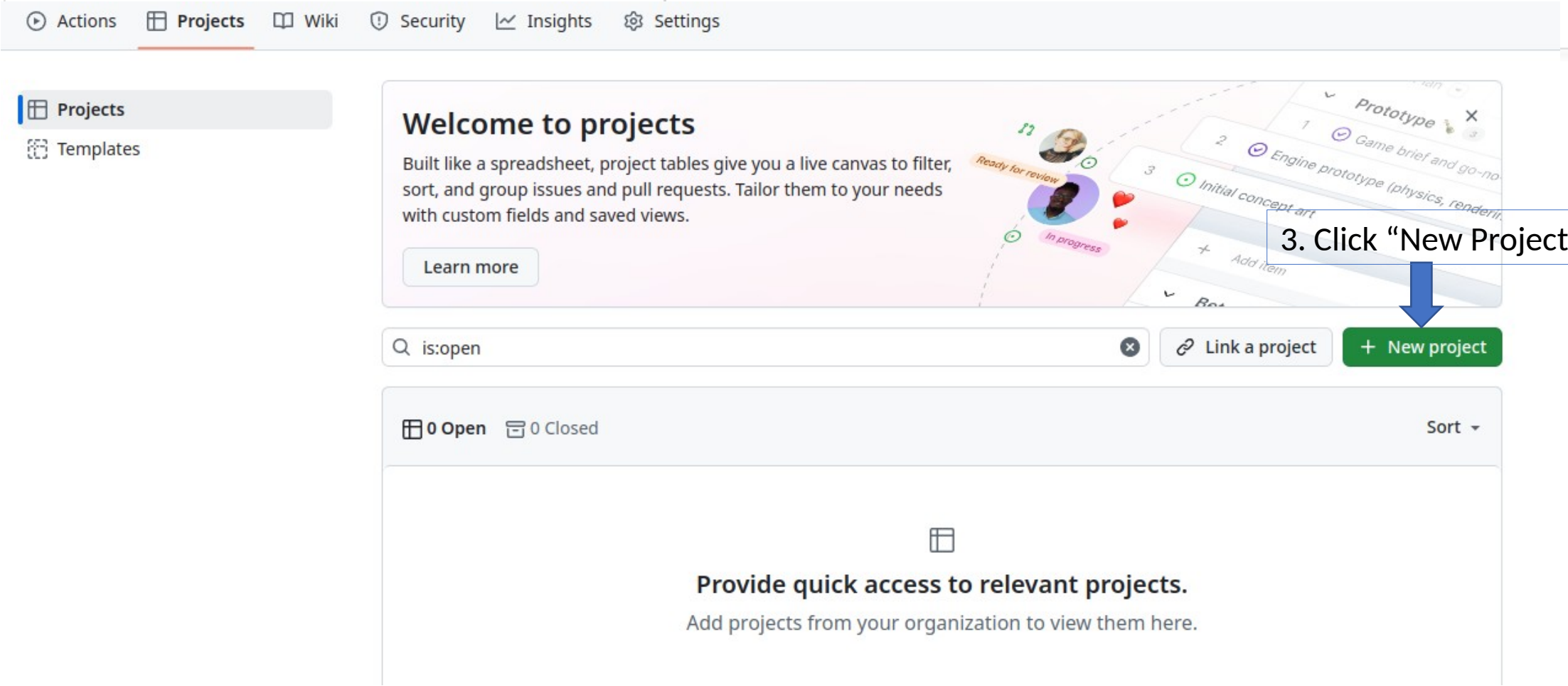
1. Work in your repository

2. Open the projects page



User Stories in GitHub

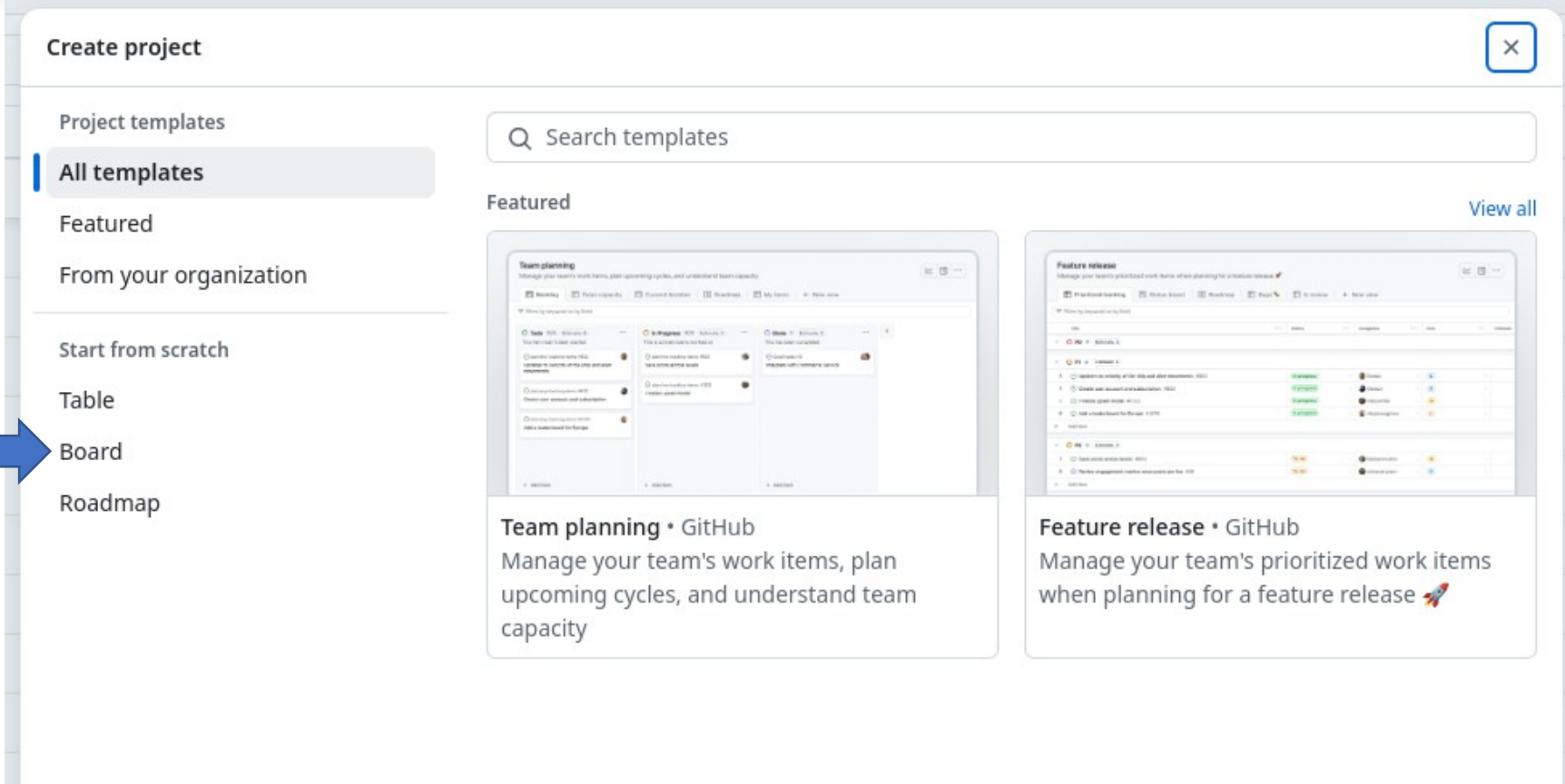
- Each user story will be its own **Projects** in GitHub



User Stories in GitHub

- Each user story will be its own **Project** in GitHub

I recommend the board template.
You can switch to table view at any time.



User Stories in GitHub

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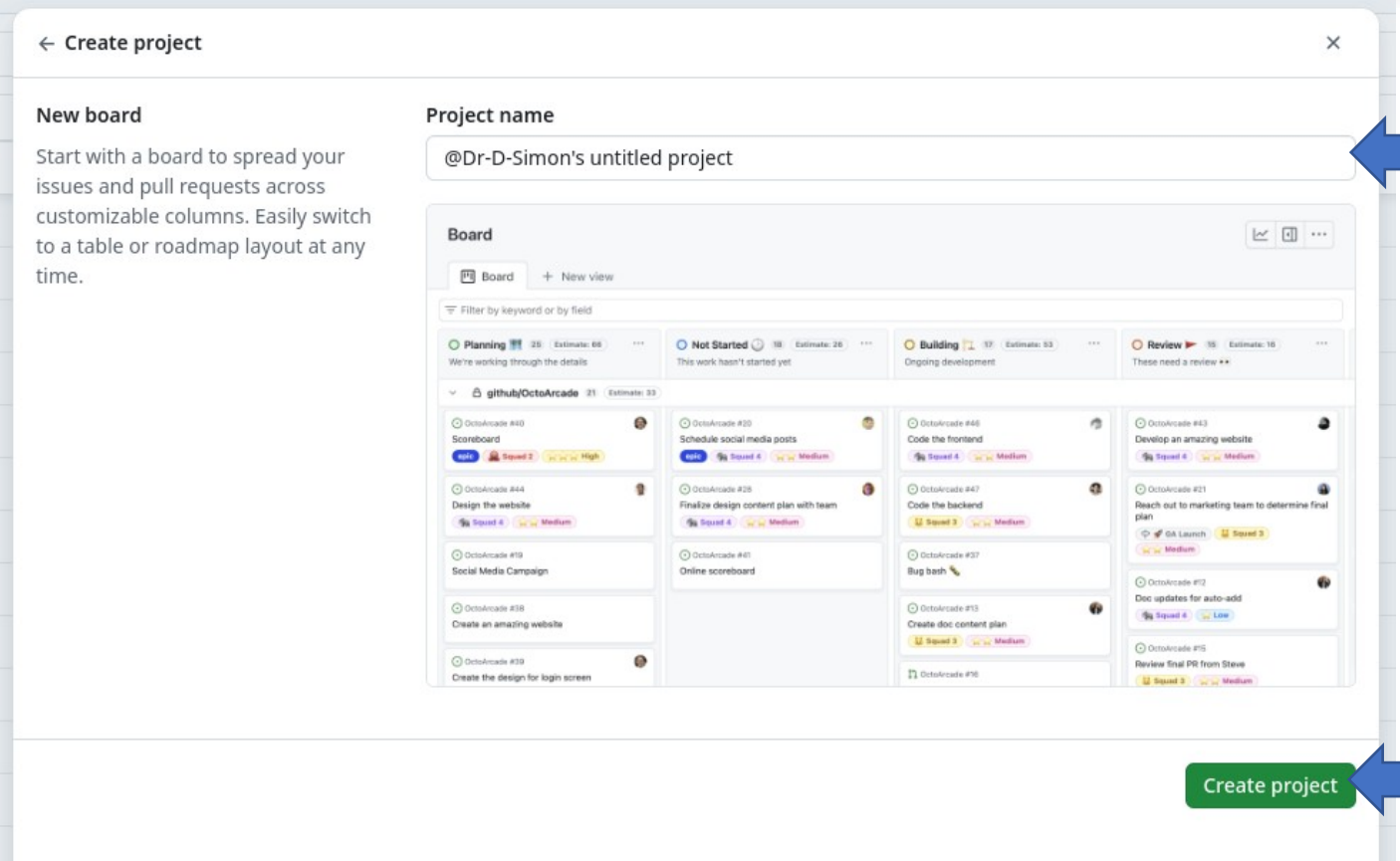


Enter your user story as the project name.

Click to complete project creation.

Choosing Milestones

Three milestones:

1. Portion of project that you think can be completed in approximately 24 hours of focused effort (leaves 4 hours for meetings, underestimates, etc.)
2. Extension of project beyond first milestone that you expect to last another 24 or so hours
3. Extension beyond second milestone that will last until at least the end of the semester

You might alter your milestones after later steps

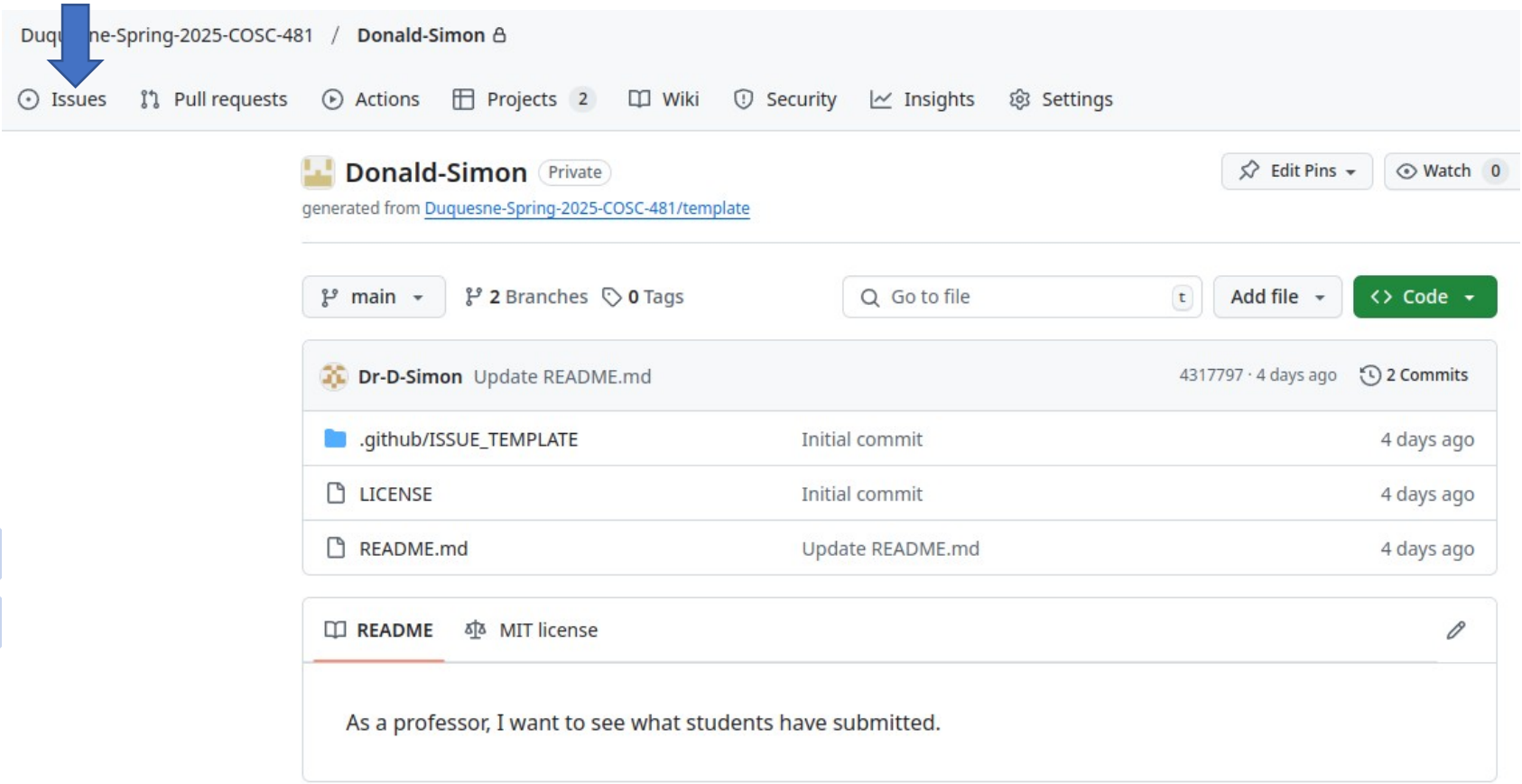
Milestones and User Stories

- Each milestone should, in addition to an overview description, consist of a list of user stories. Each user story should be part of only one milestone.
- A user story is not “done” until everything related to the user story has been completed to the required quality level:
  - Production code
  - Test code (recommend writing *before* production code)
  - User documentation, including acceptance test for the user story (list of user actions and expected system responses)
- Milestone is not achieved until all user stories are “done”

Milestones in GitHub

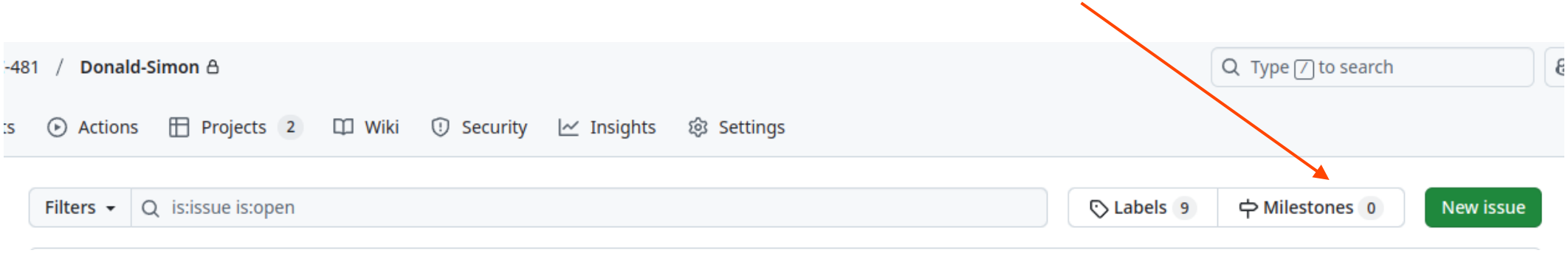
- To create Milestones:

1. Open the Issues page



Milestones in GitHub

- In the Issues area of your repository, click Milestones



- Create three milestones
  - Title: Milestone 1, Milestone 2, Milestone 3
  - Due date: February 14, March 21, April 18
  - Description: Overview description of functionality and list of user stories to be completed

Estimable Tasks

Each user story should consist of several estimable tasks

Task small enough for effort (time) to be estimated reasonably well

Development of production code for a user story often broken into multiple coding tasks for, e.g., user interface, data processing, and data storage

Include tasks for writing tests (recommend one test task for each production-coding task), user documentation including acceptance test, perhaps UI or other design tasks

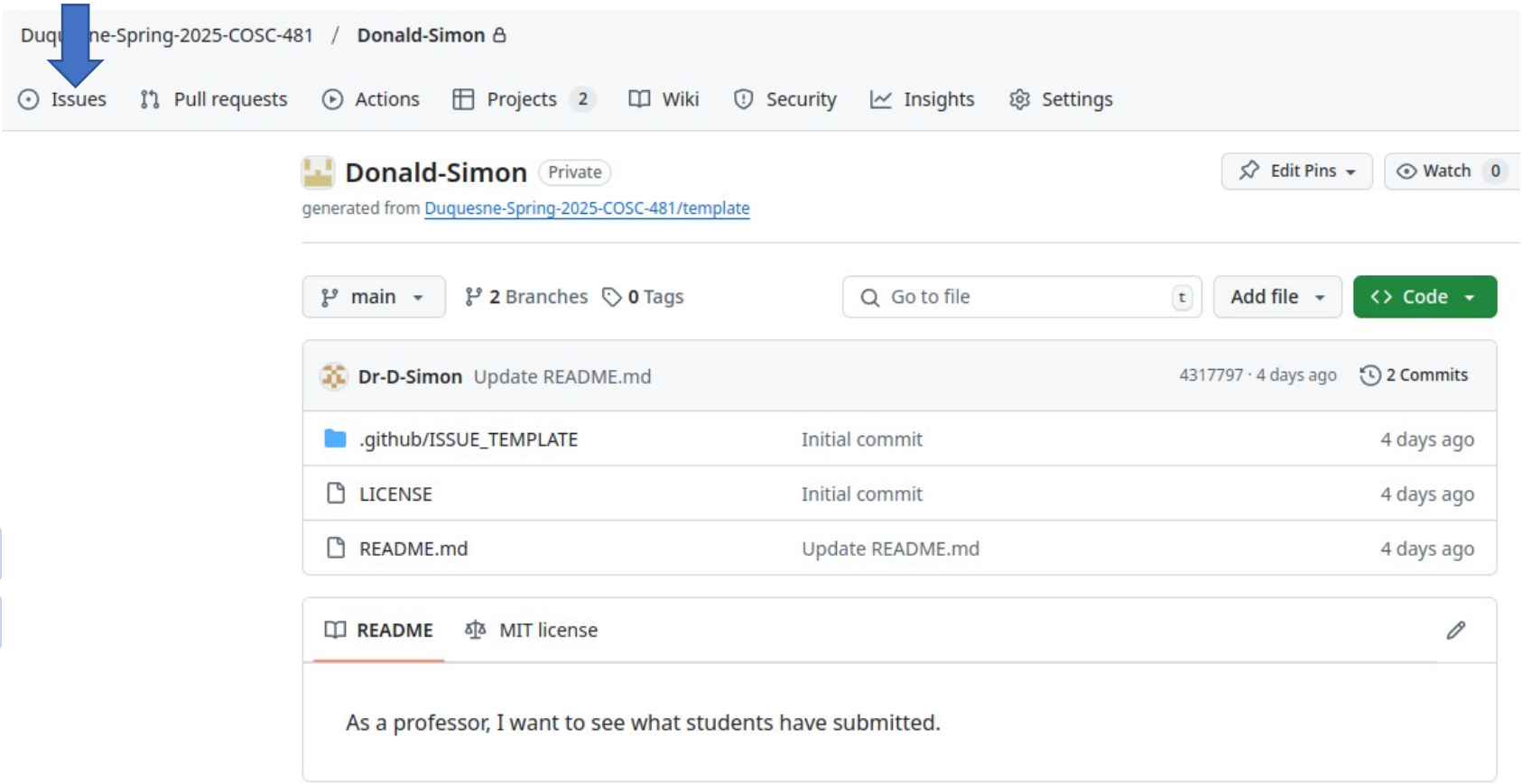
Estimable Tasks

- Estimate time required to complete each task
  - High quality, well-documented production code
  - Tests that provide complete or near-complete coverage (execute all or nearly all lines of production code)
  - Good-quality user documentation and acceptance tests
- You might find that your estimate of the total effort for Milestone 1 is very different when it is based on estimable tasks!
  - If so, you should adjust Milestone 1 by adding/removing user stories accordingly so that milestone consists of about 24 hours of effort.
  - Result of reaching milestone should be “potentially deliverable product”: Quality test/production code and documentation for milestone’s user stories.

Tasks in GitHub

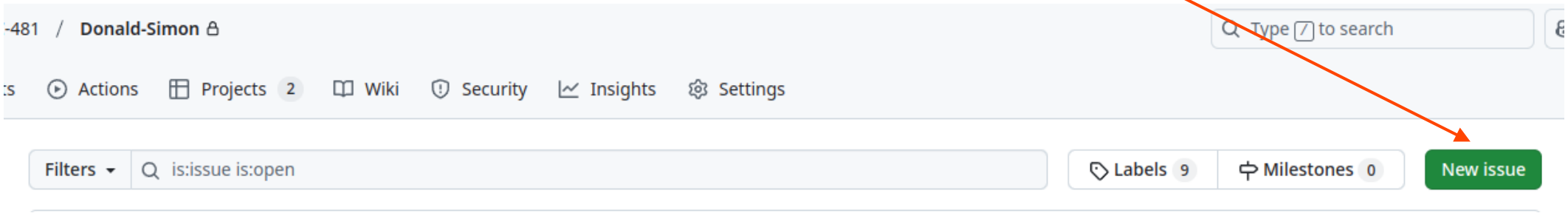
- To create Tasks:

1. Open the Issues page

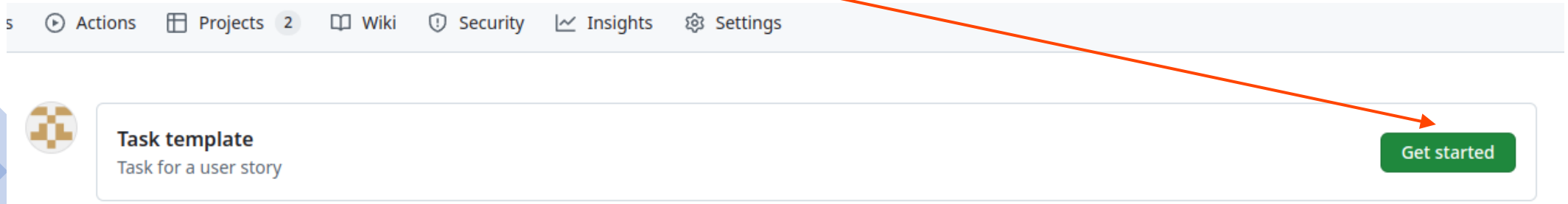


Milestones in GitHub

- In the Issues area of your repository, click “New Issue”



- Click on “Get Started”



Tasks in GitHub

Do not remove the ** ** lines. Everything else can be modified. Click *Preview* to see how it will look.

Task details as needed

Initial time estimate

0 initially

Copy of estimate initially

Issue: Task template

Task for a user story. If this doesn't look right, choose a different type.

Add a title

Title

Add a description

Write | Preview

Detailed task description.
If the task is not obvious from the issue title, provide more detail here.

Initial estimate of time required to complete this task.
Initial estimate, in hours, of the time required to complete this task. This value should not be changed.

Actual time spent on the task (so far).
Total number of hours spent on this task. This value should be updated at the end of each session working on the task.

Current estimate of time remaining to complete this task.
Estimate, in hours, of the work left to do to complete this task. This value should be updated at the end of each session working on the task.

Markdown is supported | Paste, drop, or click to add files

Assignees
No one—assign yourself

Labels
None yet

Projects
None yet

Milestone
No milestone

Development
Shows branches and pull requests linked to this issue.

Helpful resources
GitHub Community Guidelines

Submit new issue

Phrase describing task

Click gears to add task to both the correct project/user story and to Milestone 1.

Click when finished with task.

Other Week 1 Assignment Elements

- Email me at simond@duq.edu when you have completed the project elements above
- Read through Learn Git and GitHub without any code! Tutorial
- As time allows, become familiar with a set of coding standards from a reputable source for the language(s) you plan to use, for instance
  - Java: https://www.oracle.com/java/technologies/javase/codeconventions-contents.html
  - Python: https://www.python.org/dev/peps/pep-0008/
  - JavaScript: https://developer.mozilla.org/en-US/docs/MDN/Guidelines/Code_guidelines/JavaScript
- As time allows, choose a subset of language standards that you will follow and document them at your repository (in your repository Wiki, or upload a PDF or other file to branch1 in the Code area).
- Record the time spent outside of class this week at Canvas Assignment *1*.
  - I will consider 7 hours per week the average effort. Significantly less than this will likely lose "consistent effort" points.
  - If you have fewer than 100 hours at the end of the semester, you will not complete the course this semester (receive grade of F or I).